

Modeling Advection-Diffusion w/ Finite Difference Method

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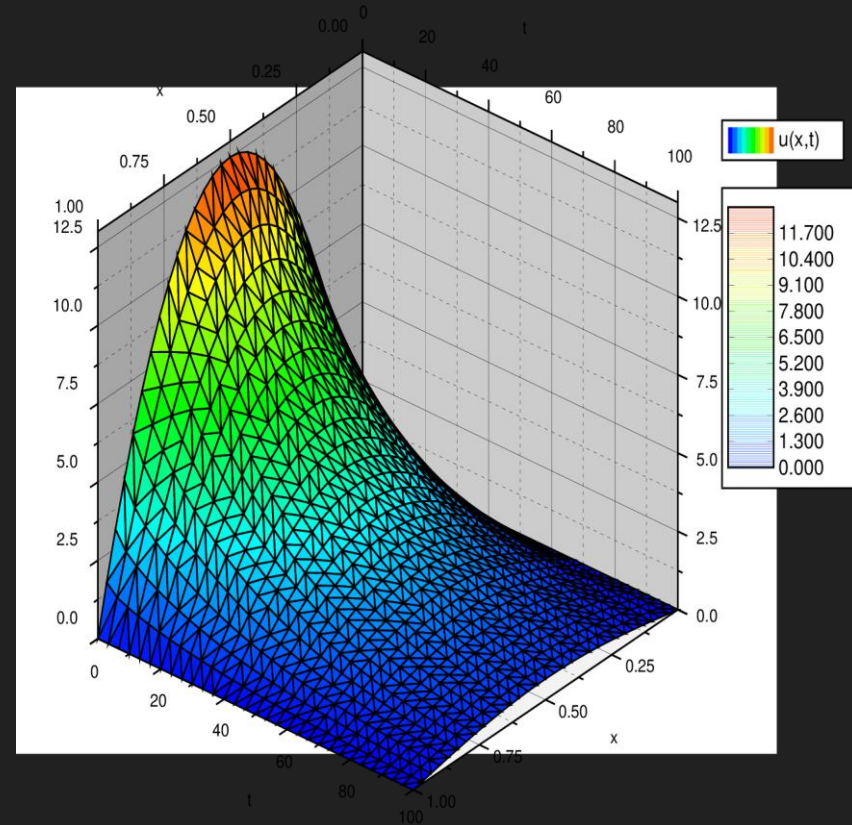
Why Advection-Diffusion?

- Wanted to be able to model the movements of forest fire using PDEs
- Proved to be way over my head
- Led me to look into natural characteristics in fluid dynamics which are also crucial in how fire spreads
- Brought me to advection and diffusion, proving to be great eq.'s for me to learn FDM using Maple!



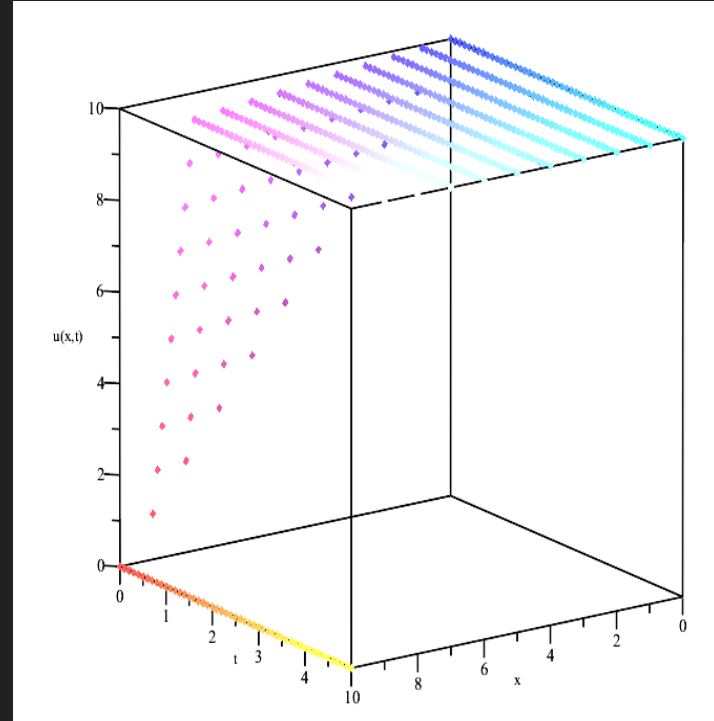
Partial Differential Equation (PDE):

- A PDE is an equation which involves a multivariable function along with one or more of its partial derivatives
- Heat Equation: $\partial u / \partial t = \alpha \partial^2 u / \partial x^2$ (where α is conductivity/(specific heat*density))
- The heat equation is first order time and second order position
- Since its creation, the heat equation has been fundamental to mathematics and developing science
- Here's a 3D plot of the equation:



Finite Difference Method (FDM):

- FDM is a numerical technique used to approximate solutions/derivatives to differential equations by discretizing them.
- Take your governing equation
(ex. $y'' - y' - y = 0$)
- Set initial conditions/boundaries
(ex. $y(0,t) = 0$, $y(x_{\text{max}},t) = 10$, $y(x,0) = f(x)$)
- Approximate your derivatives using forward, backward, or central difference method



Forward, Backward, and Central Difference Method:

- Using a Taylor series expansion, the first derivative can be approximated as:
- Forward Difference: $\partial u / \partial x \approx (u(x + \Delta x) - u(x)) / \Delta x$, x exists at $(x + \Delta x) / 2$
- Backward Difference: $\partial u / \partial x \approx (u(x) - u(x - \Delta x)) / \Delta x$, x exists at $(x - \Delta x) / 2$
- Central Difference: $\partial u / \partial x \approx (u(x + \Delta x) - u(x - \Delta x)) / 2\Delta x$, x exists at x
- This method is more accurate than the forward or backward difference since it uses a larger spatial range under u .

FDM Continued...

- Discretize using i,j notation (so instead $y(x, t) = y_{i,j}$, where $i \in 0, \dots, m$ and $j \in 0, \dots, n$)
- Create a scheme to solve for the unknown $y_{i,j+1} = y_{i,j} - k^2(y_{i+1,j} - y_{i,j})$
- Create a matrix based on number of spatial steps (m) and time steps (n) (mxn matrix)
- Insert boundary conditions into matrix and let the scheme do the rest of the work for the unknown
- Use finalized matrix to help plot the FDM solution on a 3D graph
- Preferable to use some sort of software to do the last two steps

Advection PDE:

- A very important natural phenomenon associated with fire is advection.
- Advection is the transport of a substance such as smoke or particles by the motion of a fluid (liquid or gas).
- Advection describes how the substance moves with the flow of the medium
- So in the case of a fire, it is combustion gases and particles being advected
- $-v\partial u/\partial x = \partial u/\partial t$ (typical linear advection PDE)

Convection vs. Advection:

- Convection is the movement of a fluid, in response to temperature

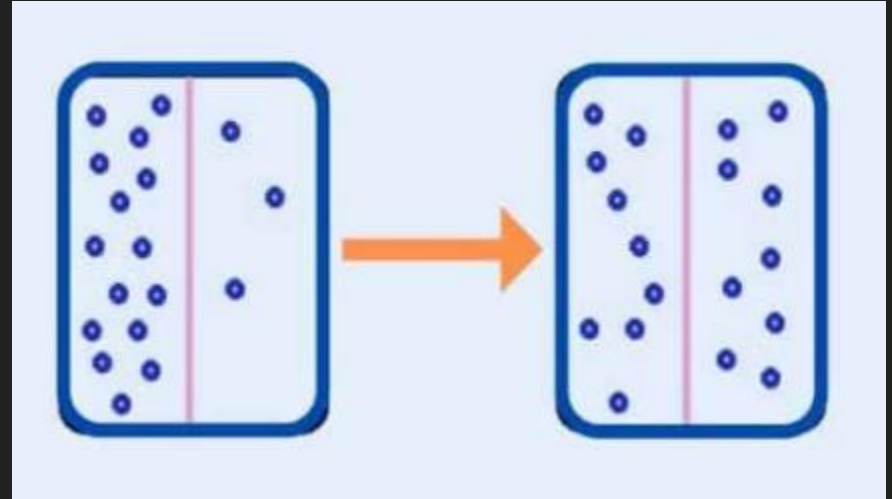


- Advection is transport of the substance within the fluid



What is Diffusion?

- Refers to the movement of particles from an area of high concentration to low concentration
- Leads to equal distribution of particles through a system



Diffusion PDE:

- Can be easily confused with advection
- Looks deeper to the molecular level where agitation and/or small-scale turbulent or chaotic motions (wind) act to move substances randomly with respect to the mean motion of fluid
- In this case combustion gases and particles are also being diffused as a result of airflow
- $\lambda \partial^2 u / \partial x^2 = \partial u / \partial t$ (diffusion PDE)

Advection-Diffusion Equation:

- What makes advection and diffusion convenient is we can combine them to make one PDE!
- They also both affect the same substances as a result of airflow! (particles and combustion gases)
- Since both PDEs are equal to $\partial u / \partial t$, all we have to do is add a diffusion term to the end of the advection equation:

$$-v\partial u / \partial x + \lambda \partial^2 u / \partial x^2 = \partial u / \partial t$$

Set Initial Conditions/Boundaries:

- $u(0, t) = 0$
- $u(N, t) = 0$
- $v(x, t) = 1$
- $\lambda = 0.25$
- $u(x, 0) = A \sin(wx)$ where $A = 1$, and $w = 1$ in this case
- This is because $u(x, t) = A \sin[\omega(x - ct)] e^{\mu \omega^2 t}$
- This will give us a much more realistic solution to our advection-diffusion PDE because flame theory is heavily involved with fluid dynamics

Approximate Derivatives Using Central Difference:

PDE:
$$-v \partial u / \partial x + \lambda \partial^2 u / \partial x^2 = \partial u / \partial t$$

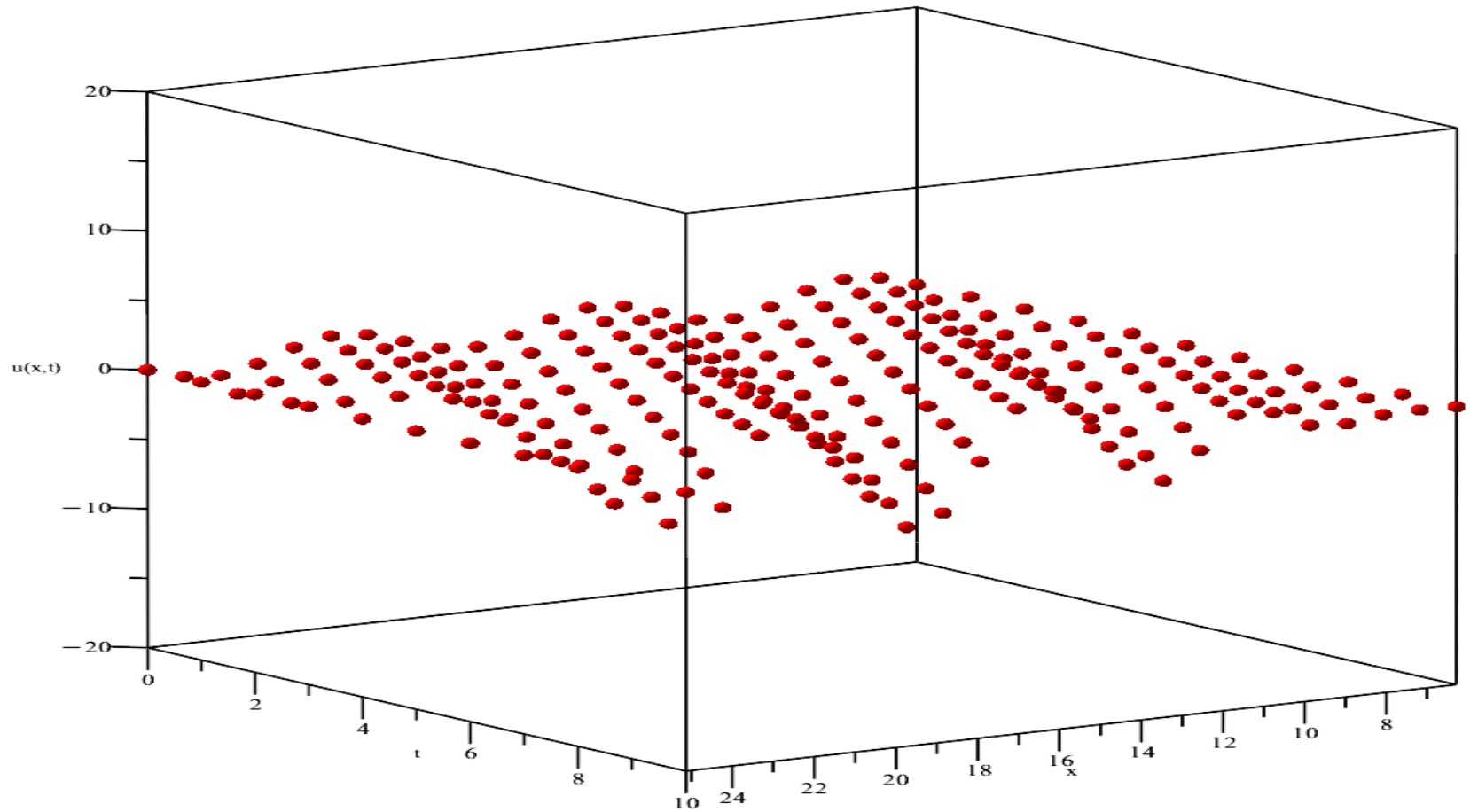
PDE Approximation:

$$-\frac{u(x,t+k)-u(x,t)}{k} - v \frac{u(x+h,t)-u(x-h,t)}{2h} + \lambda \frac{u(x+h,t)-2u(x,t)+u(x-h,t)}{h^2} = 0$$

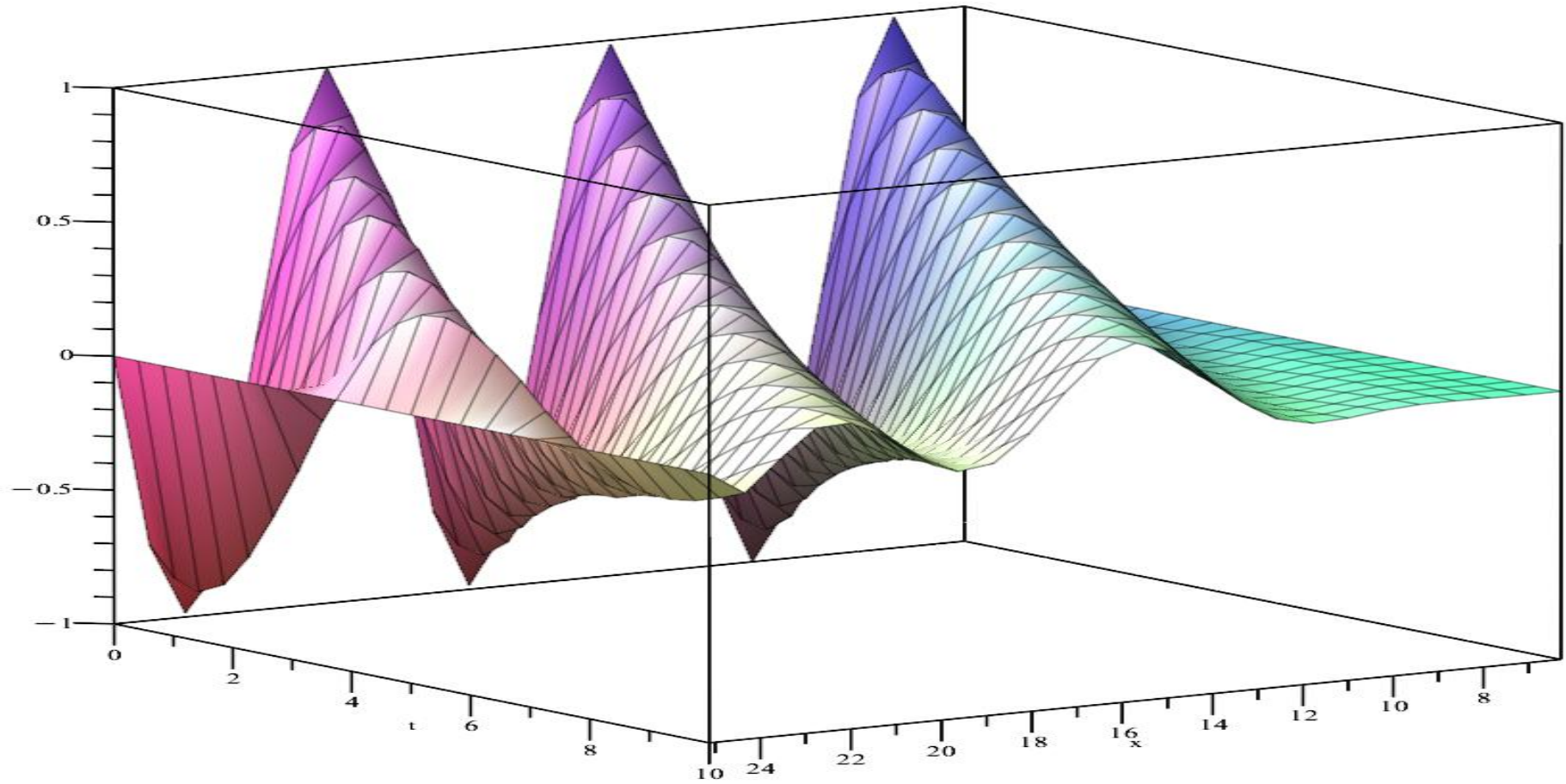
where $k = \Delta t$, $h = \Delta x$

Scheme:
$$u_{i,j+1} = u_{i,j} - vk \frac{u_{i+1,j} - u_{i-1,j}}{2h} + \lambda k \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2}$$

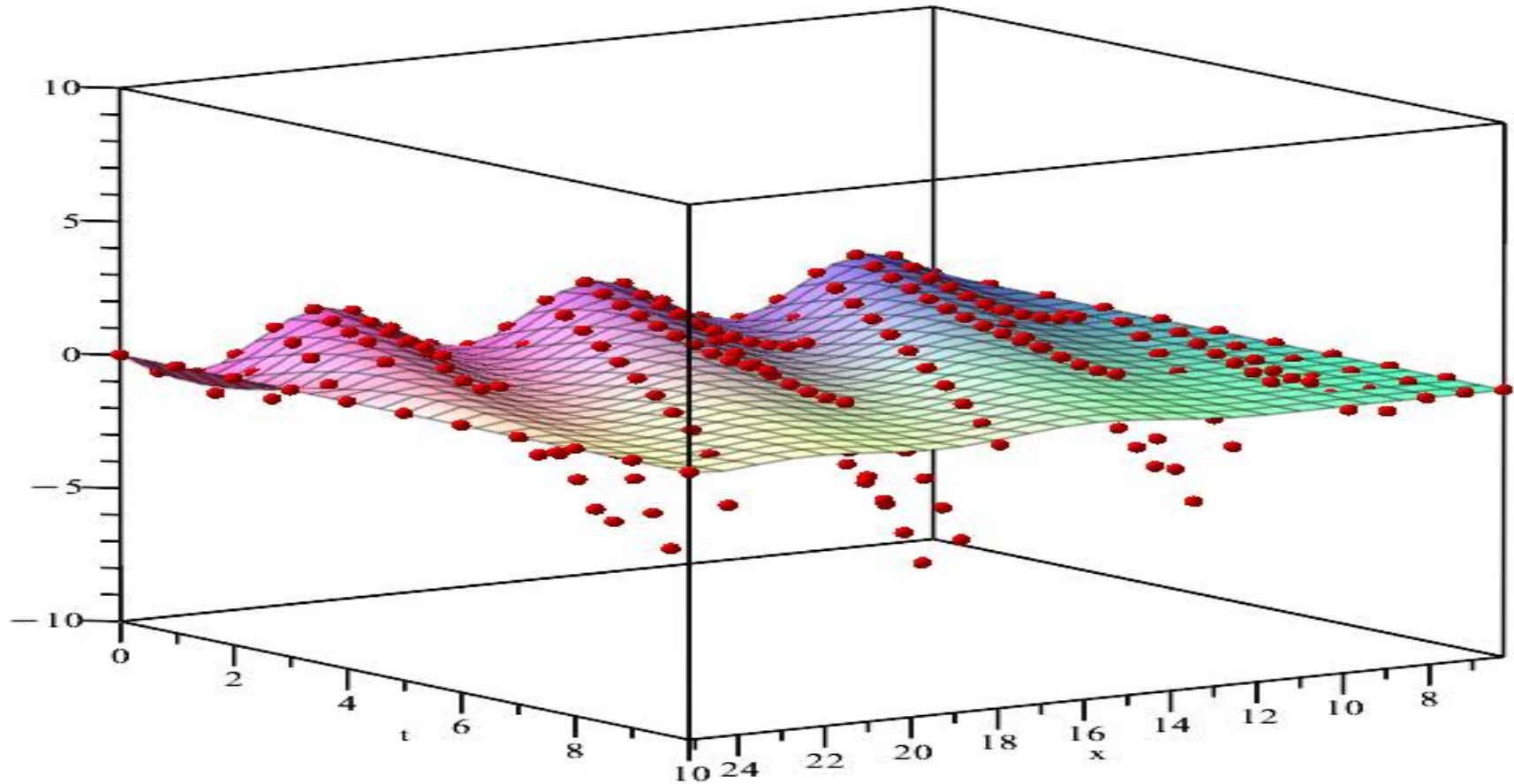
3D Plot (FDM):



3D Plot using pdesolve() (Maple function for PDEs):



FDM and pdsolve() Plots Overlaid:



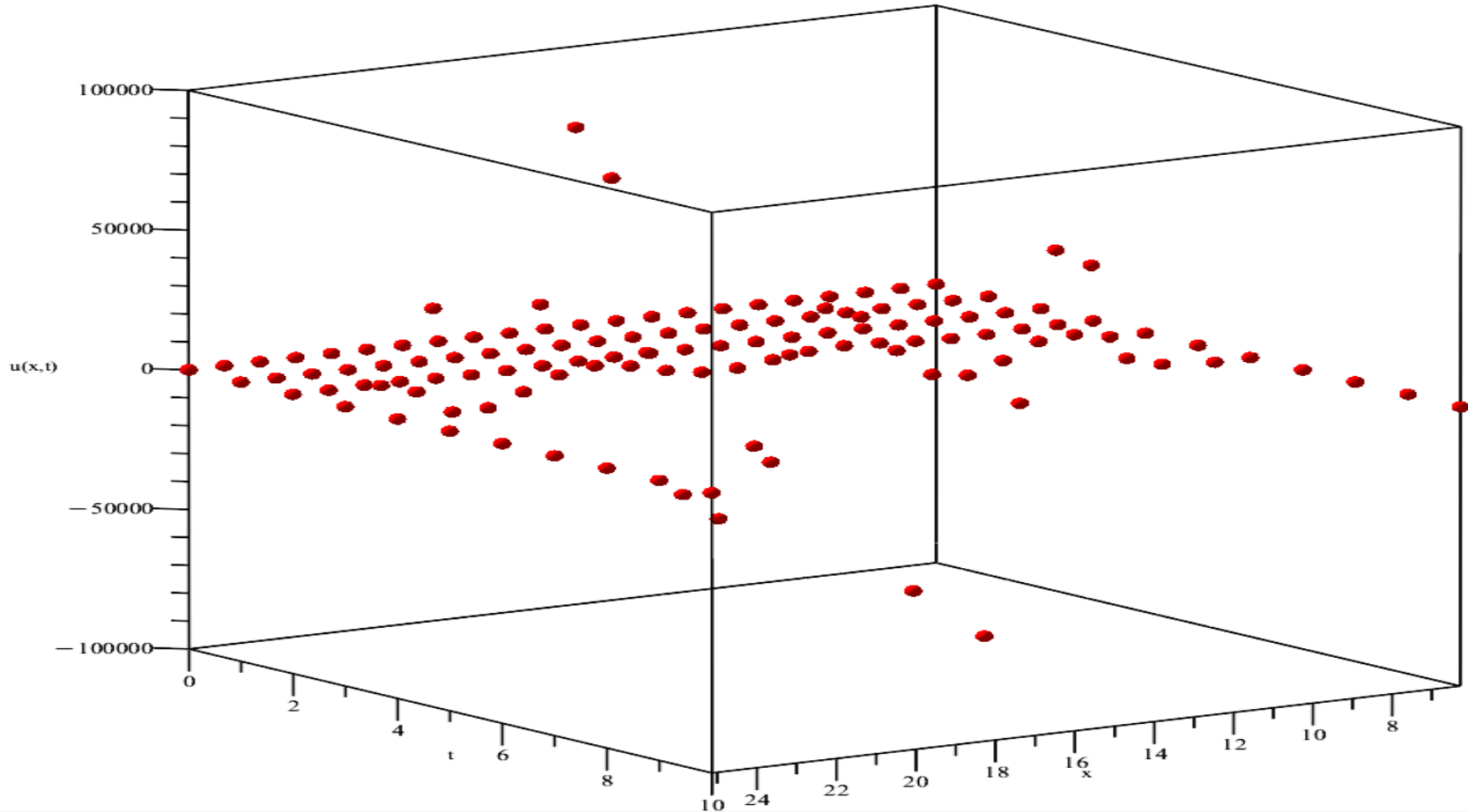
Velocity:

- $v(x, t) = x$ (instead of just 1)

$$v_{i,j} = i$$

- This will provide us a more realistic model knowing that velocity can and will change due to outside factors

Resulting 3D Plot from New ICs and Scheme (FDM):



3D Plot from New ICs and Scheme (Maple pdsolve()):

```
> # Create the plot3d solution surface
p1 := plot3d(u_exact,
             2*Pi..8*Pi, 5..15,
             axes=boxed,
             labels=["x", "t", "u(x,t)"],
             style=patch,
             transparency=0.0           # Make it slightly transparent to show underlying points
);
```

Error, (in plot3d) procedure expected, as ranges contain no plotting variables

HOW COULD THIS HAPPEN???

- A numerical method performed better than Maple's very own `pdsolve()`!
- `pdsolve()` excelled in providing the exact solution to the constant coefficient PDE, but struggled when $v(x, t) = x$
- FDM however, didn't seem to have a problem at all in implementing a solution to our upgraded PDE and even plotted it for us
- There was clearly diverging behavior but FDM still solved it (and quickly!)
- Intuitively this makes some sense because all we have to do in this case is implement another i or j into our scheme to start the numerical calculations

Key Takeaways (What Actually Happened):

- `pdsolve()` failed because it is a symbolic solver trying to find exact solutions using analytical techniques
- This can only work when the PDE is linear, or has constant coefficients, or can be reduced to an ODE, or has known Laplace transforms
- In this case, our PDE is linear but has a variable coefficient
- This caused our `pdsolve()` function to keep itself in a constant state of evaluation w/ no solution

Continued...

- FDM succeeds because the domain is discretized and derivatives are approximated numerically
- This allows us to solve very complex/nonlinear PDEs w/out exact solutions
- FDM doesn't falter just because the equation cannot be solved symbolically

Interpretation/Conclusions:

- Symbolic solvers are extremely fragile but produce amazing visuals
- Can be great for learning and quick insight for specific, well-behaved PDEs.
- Do not expect them to work for PDEs with variable coefficients, nonlinear terms, or irregular domains.
- Numerical methods serve to be much more flexible when considering new ICs
- Almost every practical PDE system (fluid dynamics, heat transfer, etc.) is solved numerically.
- Try `pdsolve()` for small, simple cases (w/ constant coefficients for example)
- Use FDM, or other numerical methods for general, complex PDEs.

Future Considerations:

- Higher dimensional models
(3D, add a 'y' term)
- Nonlinear and chaotic systems (other nonlinear PDEs w/ turbulence model)
- Integration with real data
- Comparison with other numerical methods such as FEM, and FVM



Sources:

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The background of the image is a scenic view of a sunset or sunrise. The sun is a small, bright, semi-circular orb positioned just above the horizon line. The sky is filled with horizontal bands of color, ranging from deep orange near the horizon to a darker, greyish-blue at the top. In the foreground, the dark silhouettes of trees and foliage are visible, creating a layered effect. The overall mood is peaceful and contemplative.

Thank you!